DS	**Session**	Hashing	**Question Information**	Level 1	Challenge 99

Problem Description

There are N chocolates denoted by array A where $A[i]$ is the length of the *i-th* chocolate.

Alice can melt each chocolate and then convert it into a chocolate whose length is any divisor of the number $A[i]$.

So, a chocolate of length $A[i]$ can be converted into X different types of chocolate where X is the count of divisors of the number $A[i]$.

So you need to count the total unordered pair of chocolates such that their X value is same.

Constraints

$1 \leq N \leq 10^5$
$1 \leq A[i] \leq 10^6$

Input Format

The first line contains an integer N as input denoting the total number of elements in the array A.
The next line contains N space-separated integers that denote the elements of the array A.

Output Format

In the output, print the total number of ways as mentioned in the statement.

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
3
2 3 4
```

EXPECTED OUTPUT

```
1
```

Test Case 2

INPUT (STDIN)

```
10
2 3 4 2 7 6 8 6 4 9
```

EXPECTED OUTPUT

```
12
```

Mandatory Test Cases

Test Case 1

KEYWORD

```
while(--N)
```

Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

8

Test Case 2

TOKEN COUNT

345

Test Case 3

NLOC

44

Code

chocolate_divisors.c

```c
#include <stdio.h>
#include <stdlib.h>

#define MAXVAL 1000005

int div_count[MAXVAL];
long long freq_div[1000];

void precompute_divisors() {
    for (int i = 1; i < MAXVAL; i++) {
        for (int j = i; j < MAXVAL; j += i) {
            div_count[j]++;
        }
    }
}

int main() {
    precompute_divisors();

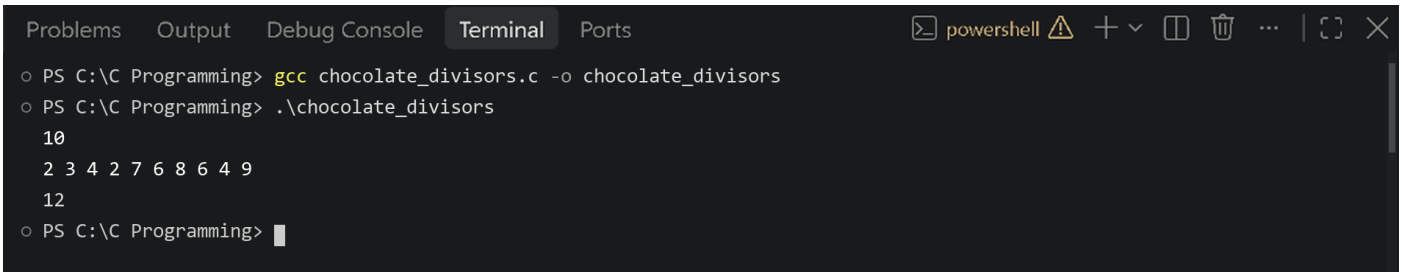
    int N;
    if (scanf("%d", &N) != 1) return 0;

    int orig_N = N;
    while(-N) {
        if (N <= 0) break;
        int val;
        scanf("%d", &val);
        freq_div[div_count[val]]++;
        N--;
    }

    long long total_pairs = 0;
    for (int d = 1; d < 1000; d++) {
        if (freq_div[d] > 1) {
            total_pairs += freq_div[d] * (freq_div[d] - 1) / 2;
        }
    }

    printf("%lld\n", total_pairs);
    return 0;
}
```

Output


Problems
Output
Debug Console
Terminal
Ports
powershell
PS C:\C Programming> gcc chocolate_divisors.c -o chocolate_divisors
PS C:\C Programming> .\chocolate_divisors
10
2 3 4 2 7 6 8 6 4 9
12
PS C:\C Programming>